

# Model for grain boundary sliding and its relevance to optimal structural superplasticity

## Part 2 – Evidence for cooperative grain/interphase boundary sliding and plane interface formation

V. V. Astanin, S. N. Faizova, and K. A. Padmanabhan

*A Zn–22 wt-%Al alloy was superplastically deformed in the optimal range to obtain experimental evidence based on scanning electron and transmission electron microscopy for mesoscopic grain/interphase boundary sliding and plane interface formation that facilitates the cooperative boundary sliding process. It was seen that such localised shear surfaces had formed in the entire specimen bulk in an interconnected manner. At boundaries that formed part of a cooperative shear surface the triple junctions had migrated to a near 180° orientation. Mesoscopic grain/interphase boundary sliding appeared to be accommodated by local boundary migration brought about by diffusion at intercrystalline boundaries and lattice dislocation motion at interphase interfaces in accordance with the requirements of the model presented in Part 1.*

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© 1996 The Institute of Materials. Manuscript received 2 June 1994; in final form 6 July 1995. Dr Astanin and Ms Faizova are at the Institute of Metals Superplasticity Problems, Russian Academy of Sciences, Ufa, Russia, and Professor Padmanabhan is at the Department of Metallurgical Engineering, Indian Institute of Technology, Madras, India.

### Introduction

Models for structural superplastic flow available in the literature<sup>1–6</sup> consider sliding along individual grain boundaries only and in many cases the analysis is two-dimensional. In recent times it has become clear that as a result of its complex spatial configuration a grain may have boundaries that are located favourably for sliding and others at which grain/interphase boundary sliding is difficult or absent.

Cooperative sliding at grain/interphase boundaries in which a number of grains slide as an entity along shear surfaces located in different regions of the bulk of the specimen has been reported.<sup>7–10</sup> These surfaces are produced by the joining together of grain boundaries that are oriented favourably to the maximum shear stress directions in the deforming specimen. Sliding of grains along surfaces has been examined by Raj and Ashby,<sup>11</sup> but their analysis involves considerable diffusional accommodation for which there is no experimental support in superplastic flow. Further, they regard diffusional accommodation as the rate controlling process while the contemporary view favours grain/interphase boundary sliding as the rate determining mechanism in superplasticity (see Part 1 (Ref. 12)).

Grain/interphase boundary sliding is a two scale phenomenon.<sup>12</sup> At the atomic level, sliding at a boundary operates on a microscopic scale. Steric hindrance makes this kind of sliding relatively ineffective in bringing about large plastic deformation without help from additional mechanisms. Such atomic scale boundary sliding is common in high temperature creep, where it leads to cavitation, triple point cracking, and premature failure. If boundary sliding is to be as extensive as that seen during superplastic flow, sliding of the order of a grain diameter and more must be present (referred to in Part 1 (Ref. 12) as 'mesoscopic sliding').

Mesoscopic sliding gives rise to a predictable long range threshold stress which has to be overcome by the applied stress, in addition to the short range internal stresses arising from microscopic sliding. It is these internal and threshold stresses that are responsible for regions I and IIa of superplastic flow<sup>4</sup> having different strain rate sensitivity indices ( $m$  values) even though both these regions are controlled by the same physical processes (see Part 4 for numerical results).

Plane interface formation needed for mesoscopic (cooperative) sliding is caused at an intercrystalline boundary by local boundary migration brought about by diffusion. (If the boundary obstacle is severe, local boundary migration may also result from non-rate controlling lattice dislocation motion as occurs at interphase interfaces (see below).) However, at an interphase interface this can only be brought about by non-rate controlling lattice dislocation motion originating at interphase boundary obstacles or regions of high curvature.<sup>12</sup>

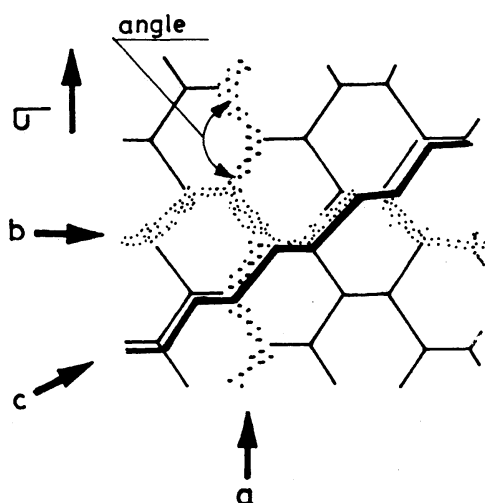
According to Karl Popper, a good theory must contain an element that can be either proved or disproved by a direct experiment, i.e. the theory should have 'the quality of falsifiability'.<sup>13</sup> In the present case the plane interface needed for mesoscopic sliding should form in the first instance and when this involves an interphase boundary, lattice dislocation motion should be a concomitant of this process. Experimental evidence concerning the formation of the interface needed for mesoscopic sliding is presented below.

### Material and experimental procedures

In order to obtain an optimal microstructure, 10 mm thick ingots of the Zn–22 wt-%Al alloy were water quenched from 623 K and during the period of spontaneous heat release from spinodal decomposition were rolled down to a thickness of  $\sim 0.6$  mm. Specimens ( $14.0 \times 5.0 \times 0.6$  mm) were prepared by filing, and annealed at 523 K for 2.5 h. A stable two phase microstructure with an average grain size of  $1.1 \mu\text{m}$  was obtained. For making quantitative measurements, marker lines along and transverse to the extension axis were introduced using diamond paste.

The specimens were tensile tested on an Instron universal testing machine at a temperature of 523 K and an initial strain rate of  $6.9 \times 10^{-3} \text{ s}^{-1}$ , which corresponded to the optimal conditions for superplasticity in this alloy.<sup>4,6</sup> To maximise the relief, a thin film of carbon was applied on the specimen surface by vacuum deposition. This film broke at the sites of intense deformation, revealing them as bands of contrast when examined in an SEM.

After a pre-determined degree of deformation, reference points were introduced into the specimen using a diamond indenter and the region between the reference points



a along tensile axis; b across tensile axis; c at 45° to tensile axis

1 Schematic diagram showing measurement of angles between adjoining grain boundaries in different directions

photographed in an SEM. A layer of two to three grains in thickness was removed by electrolytic polishing and another photograph of the same region at this depth was taken in the SEM. The two photographs were compared by superposition and the traces of cooperative grain/interphase boundary sliding bands were identified. Similar photographs were used to measure the angles between the adjoining members of the boundary chains, along the tensile direction, in the transverse direction, and at 45° to the tensile axis. The scheme of measurement is shown in Fig. 1.

To determine the structure at a high magnification, thin foils were prepared at  $-60^{\circ}\text{C}$  by electropolishing in a 10% solution of  $\text{HClO}_4$  in ethanol. The foils were examined in a JEM-2000 type TEM.

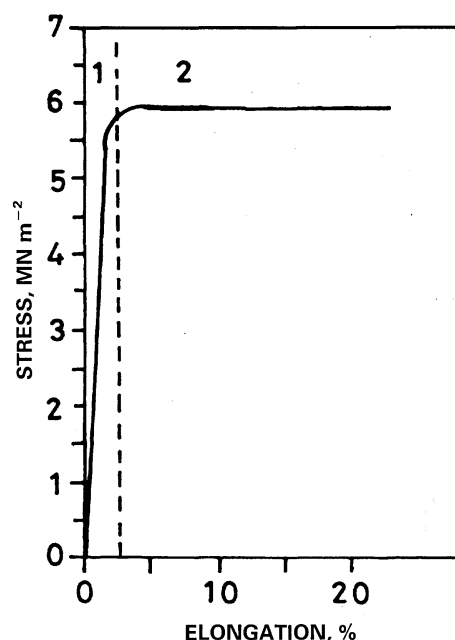
## Results

From the true stress–elongation data for the Zn–22 wt-%Al alloy (Fig. 2) and similar results for other systems reported in the literature, two regions are clearly discernible:

- (i) a region where the true stress  $\sigma$  depends strongly on the true strain  $\epsilon$
- (ii) a region where the strain dependence of stress is weak (the region of stable flow). The mechanical response in these two regions correlated well with the details of boundary shear band formation.

The microstructural changes present in the initial region of deformation in which the stress was strongly dependent on the strain were described earlier.<sup>9,14</sup> In this range, flow commenced at very low stresses and no visible elastic ( $\sigma \propto \epsilon$ ) deformation was present. Bands of intense shear deformation due to cooperative grain/interphase boundary sliding were seen to develop (see below). The rapid rise in the flow stress correlated well with the increase in the number and dimensions of these directional, cooperatively slid bands. This process continued until a steady state (in terms of both the number and the dimensions of these bands) was reached, beyond which point the strain dependence of the flow stress became very weak.

The formation of each band of mesoscopic sliding involves overcoming a long range threshold stress (Part 1). So, it is understandable that the stress required to enforce deformation increased with strain until these bands reached their steady state number and dimensions, i.e. until that



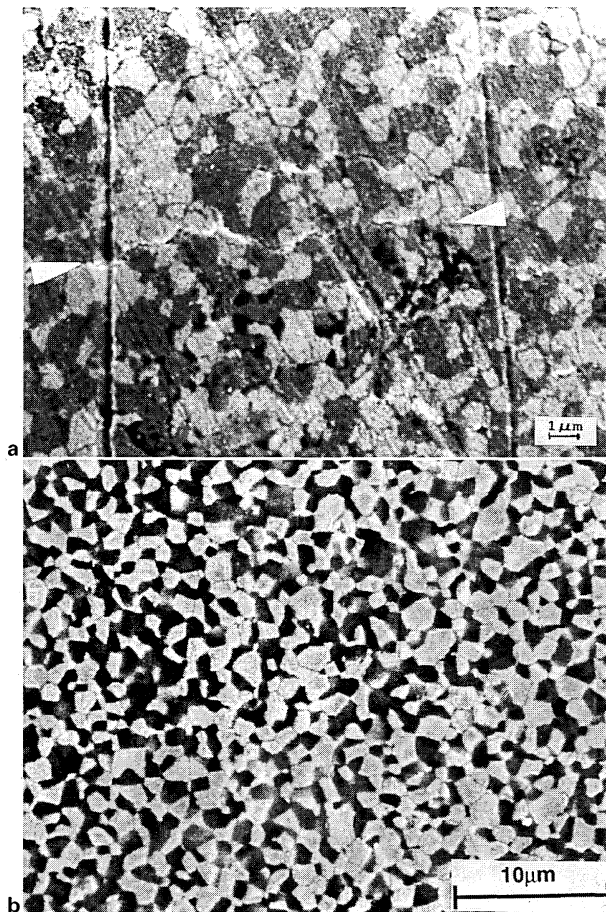
2 Dependence of flow stress on specimen elongation in initial stage of extension

stage at which a continuous network in three dimensions of the cooperatively slid bands had formed.<sup>14</sup> A rigorous analysis for this unsteady region of flow would be extremely complex and has yet to be given. However, a simplified quantitative description was presented some time ago.<sup>15</sup>

At an elongation of 3.0% (corresponding to the end of stage (i) above) at the optimal strain rate, thin ( $0.1\ \mu\text{m}$ ), winding shear bands  $15\text{--}20\ \mu\text{m}$  long began to appear along grain boundaries at a distance of  $5\text{--}8\ \mu\text{m}$  apart (Fig. 3a). The bands were arranged across the specimen from one of its edges to the other. They were not distributed homogeneously in the sheet plane but were concentrated in certain areas. At places in the specimen where such concentrations of shear bands were present, macroscopic necks were seen. The displacements of marker lines crossing these bands revealed evidence for the relative shear of one group of grains with respect to another along the direction of the bands, i.e. the shear bands were the result of cooperative grain/interphase boundary sliding.<sup>8–10</sup>

To establish that the above changes were indeed due to deformation, an SEM image of an undeformed sample is shown in Fig. 3b for comparison. The shear band formed in Fig. 3a by cooperative boundary sliding appears to be normal to the stress axis. That this is not really so can be seen from the schematic shown in Fig. 3c. What is seen in Fig. 3a is the trace in the plane bounded by the specimen length and breadth of the cooperatively slid boundary that is at approximately  $45^{\circ}$  to the stress axis in the plane bounded by the specimen length and thickness. The appearance of the cooperatively slid boundaries in three dimensions was described earlier.<sup>9,14</sup>

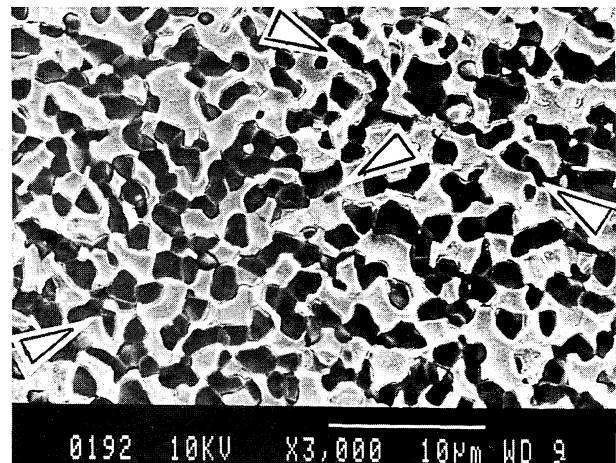
Evidence of cooperative boundary shear band formation after an elongation of 200% was observed even after removal of a surface layer by electropolishing, which implied that along these lines of cooperative boundary sliding the angles between adjoining boundaries differed significantly from the equilibrium value of  $120^{\circ}$  and in fact approached  $180^{\circ}$ , i.e. the boundaries had become less angular or a nearly straightened chain had formed as a result of cooperative boundary sliding (Fig. 4). Data on the angles at triple junctions measured in different directions are shown in Fig. 5 while their mean values and those of the 'saw tooth' parameters for mesoscopic sliding, which is described more fully in the following section, are given in



3 a Boundary shear relief in initial stage of specimen extension: arrows show bands of cooperative grain boundary shear, tensile axis is vertical; b SEM picture of undeformed specimen for comparison; c schematic diagram illustrating why shear band in a appears normal to stress axis: CGBS = cooperative grain boundary shear

Table 1. It was also evident that after an elongation of 200% chains of  $\beta$  phase grains had formed as part of the cooperative grain/interphase boundary sliding bands (Fig. 4).

Figure 6a shows a TEM image of a specimen deformed up to the onset of steady state flow. A cooperative grain/interphase boundary sliding band is present along one of the diagonals of this figure. The location where a triple junction has migrated to form a near plane interface is indicated by an arrow. Where a  $\beta$  phase grain interrupted



4 Microstructure after 200% elongation and removal of surface layer by electropolishing: arrows indicate bands of cooperative grain/interphase boundary shear that are identified by superposition of photographs of same region before and after electropolishing; tensile axis is vertical

plane interface formation (i.e. an interphase interface was involved), the process continued by dislocation motion through the  $\beta$  grain which appears to have given rise to a low angle boundary within the  $\beta$  grain (indicated by a concentration of dislocations) for continuing cooperative sliding. Grain boundaries that formed part of a cooperatively slid boundary band often looked like 'fringes' while the other boundaries had a smooth appearance.

## Discussion

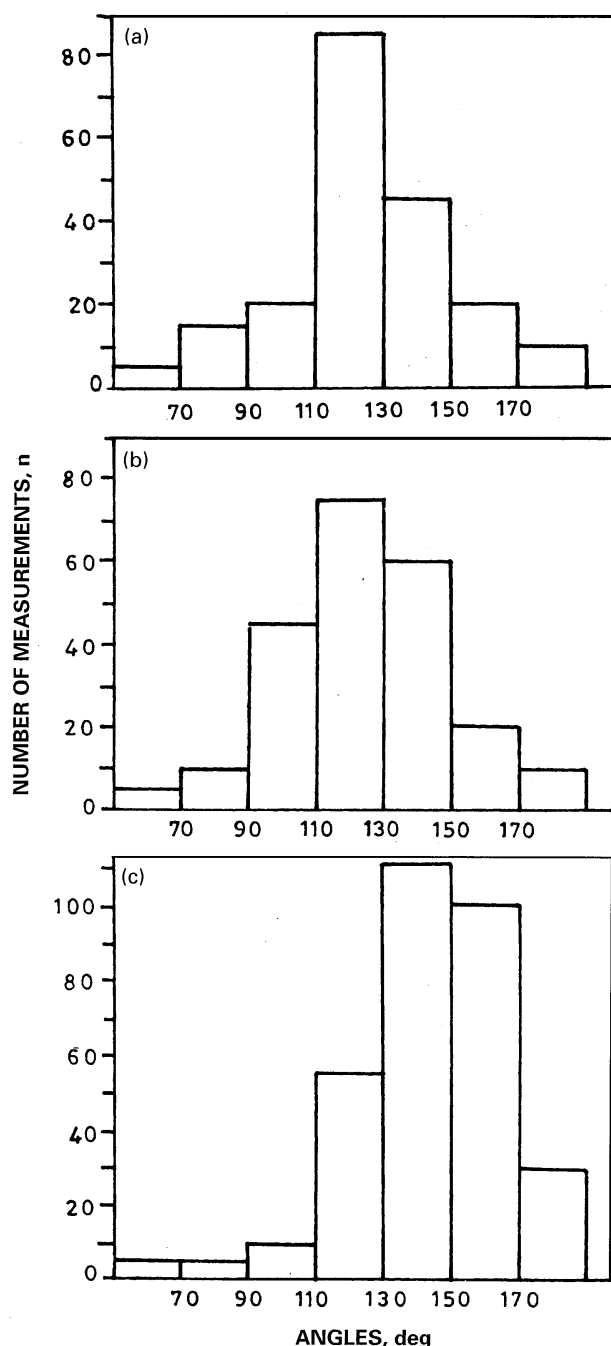
The present results concerning cooperative (mesoscopic) grain/interphase boundary sliding are best understood in terms of Fig. 1 of Part 1.<sup>12</sup> In that analysis the polycrystalline structure on a mesoscopic scale is described as a cubic dense packing of equisized spheres. The walls of the Wigner-Seitz cell containing one sphere represent the grain boundaries. This corresponds to a face centred cubic structure on a mesoscopic scale. Since roughness is lowest on the  $\{111\}$  planes, these are assumed to be the glide planes for long range (mesoscopic) sliding. The surface of a  $\{111\}$  plane in the mesoscopic structure consists of a regular arrangement of peaks and troughs whose height, regardless of grain shape, will vary as  $L$ , the average grain size. The problem of plane interface formation along the mesoscopic  $\{111\}$  plane has been treated mathematically in Part 1.<sup>12</sup>

Initially sliding commences at boundaries that experience the largest shear stresses. Here again, sliding will be easier when the grain boundary misfit is greater. In Zn-22 wt-%Al and some other two phase alloys, sliding at interphase interfaces is easier than along intercrystalline bound-

Table 1 Average values of angles at triple junctions and 'saw tooth' parameters: for definitions of  $s$  and  $h$ : see Fig. 6b

| Direction<br>(see Fig. 1) | Average angle at<br>triple junctions,<br>deg | $h$ , $\mu\text{m}$ | $s$ , $\mu\text{m}$ |
|---------------------------|----------------------------------------------|---------------------|---------------------|
| a                         | 122                                          | 0.43                | 1.56                |
| b                         | 123                                          | 0.40                | 1.43                |
| c                         | 142                                          | 0.30                | 1.62                |

\* Represents a region near the grain boundary disturbed by deformation and should not be confused with grain boundary width.

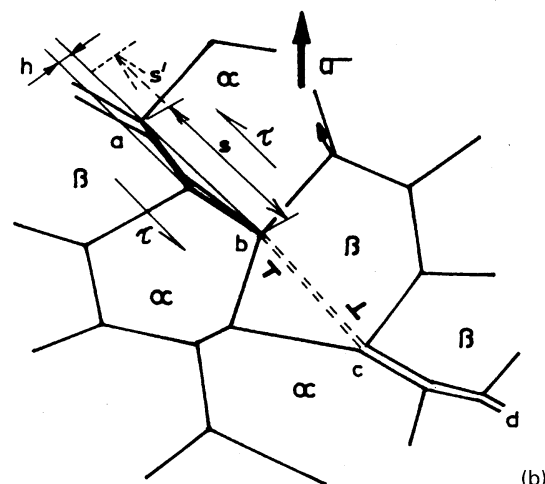
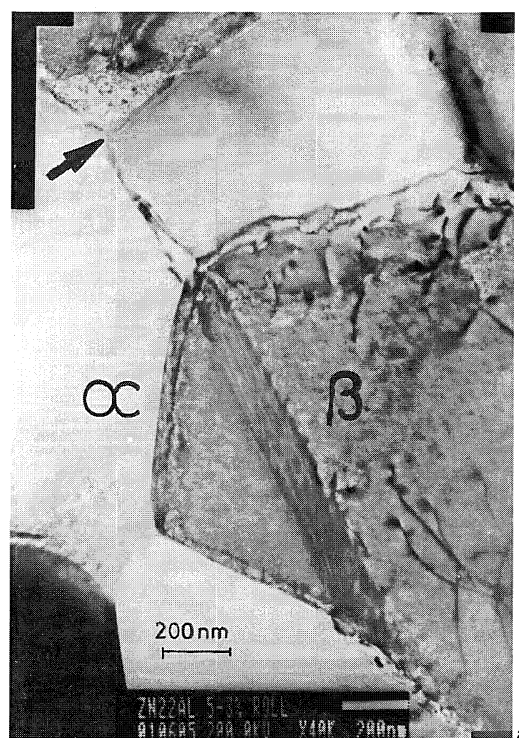


a along tensile axis; b across tensile axis; c at 45° to tensile axis

##### 5 Statistical data on angles present at triple junctions measured in different directions

aries.<sup>16-18</sup> This is verified in the present study where it has been seen above that  $\beta$  phase grains are aligned along the cooperative boundary sliding bands (Fig. 4).

According to Kaibyshev and co-workers,<sup>6,18</sup> lattice dislocations that originate at a grain boundary traverse the grain and are absorbed at an opposite boundary, making it of a 'more non-equilibrium' nature. This process makes sliding easier and faster ('stimulated grain boundary sliding'). As the number of such dislocations can be expected to increase with stress (strain rate), it may be argued that superplasticity will be more significant in region II than in region I. While the evidence for dislocation motion commencing from a boundary and traversing the grain to become absorbed at an opposite boundary found in the present experiments (Fig. 6a) can be reconciled with the above view, the plane interface formation needed to



6 a Transmission electron micrograph corresponding to 3% elongation: arrow indicates where triple junction has migrated to form near plane interface; b schematic diagram for formation in initial stage of extension of cooperatively slid grain/interphase boundary band

cause cooperative boundary sliding observed here is not predicted by the analysis of Kaibyshev *et al.*<sup>6,18</sup> Moreover, the concept of a change in the degree of the non-equilibrium nature of a boundary when a lattice dislocation enters it, has yet to be quantified.

In the analyses by Ashby and Verrall<sup>1</sup> and Raj and Ashby<sup>11</sup> the starting two-dimensional array consists of regular hexagons of uniform size. However, in the present experiments at a number of sites the grain shape was irregular and the angle at triple junctions significantly differed from 120°. Figure 6b is a schematic representation of a region similar to that shown in Fig. 6a. Evidently, as  $s$  increases and/or  $h$  decreases, sliding between a pair of grains can be more easily transferred to an adjoining pair of grains to cause cooperative grain/interphase boundary sliding. However, a point such as  $b$  in Fig. 6b would correspond to the grain boundary discontinuity involving an interphase interface discussed in Part 1.<sup>12</sup> To continue cooperative sliding and plane interface formation beyond

a-b to a-c, lattice dislocation motion will have to commence at the boundary at b, traverse the  $\beta$  grain, and become absorbed at the boundary at c. Such a process is indeed seen in Fig. 6a. It is pertinent to note at this stage that using TEM it is not possible to determine if the lattice dislocation motion observed in Fig. 6a is non-rate controlling. This can only be determined by texture results based on a study of true crystallite orientation distribution functions. It has already been shown that under optimal conditions of superplastic flow, texture intensity decreases continuously as a result of boundary sliding and grain rotation, and there is no evidence for crystallographic slip acting as an independent rate controlling mechanism (see, for example, Fig. 10 of Ref. 19). Thus it can be concluded that what is observed in Fig. 6a is evidence for plane interface formation involving an interphase interface by non-rate controlling dislocation motion through the  $\beta$  grain, which is in accordance with the requirement of the model presented in Part 1.<sup>12</sup> In other words, the model presented in Part 1 passes the test based on 'the quality of falsifiability'<sup>13</sup> (see 'Introduction' above).

In contrast, the curvature present at an intercrystalline boundary is removed by local boundary migration caused by diffusion. (If the boundary obstacle is severe, this can also be achieved by dislocation motion, as described above.) The energy needed to form the plane interface can be estimated using equations (33) and (34) and the relationship  $\Delta W_3 = \Delta W_2/2 = \Delta W_1$ , as defined in Part 1. Likewise the long range threshold stress to be overcome before a plane interface can form can be obtained from equations (36)–(38). It is also clear that in Fig. 6b the ground area indicated in terms of  $s$  in cross-section varies as  $L^2$  and the height  $h$  to be removed before a plane interface can be formed varies as  $L$ , where  $L$  is the grain size.

Plane interface formation, as described above, will also be present in section d-c of Fig. 6b. As in the model,<sup>12</sup> this process can take place simultaneously in a number of regions of the deforming specimen, when the plane interfaces a-c and d-c meet, the stress concentration at c will be reduced by a local boundary smoothing (migration) process brought about by diffusion. Then, as experimentally observed, the surfaces along which cooperative boundary sliding is present will acquire a wavy or curved appearance (Fig. 3).

From the present results it is also clear that the grain boundaries in which the dissociation of the absorbed lattice dislocations is incomplete acquire a fringelike appearance.

In Fig. 6a, one of the grains meeting at a triple junction (indicated by an arrow) has migrated so as to extend the plane interface causing cooperative boundary sliding. The presence of such local boundary migration (caused by diffusion at an intercrystalline boundary and non-rate controlling lattice dislocation motion at an interphase interface, as described in Part 1<sup>12</sup>), that leads to cooperative boundary sliding on an extensive scale is a unique feature of superplastic flow which is absent in high temperature creep in which atomic scale grain boundary sliding is also present.

The limitations of the Ashby-Verrall<sup>1</sup> and Raj-Ashby<sup>11</sup> analyses, both of which treat diffusional flow as the rate controlling process, have already been discussed.<sup>4,6</sup> Of special relevance are the two results presented in Fig. 6a, neither of which can be accounted for by the above diffusion based models, i.e. plane interface formation involving several grain/interphase boundaries which makes them behave as an entity, and the evidence for accommodation of sliding at an interphase interface by dislocation motion to cause cooperative boundary sliding in the optimal range of superplastic flow. These observations are, however, essential features of the model presented in Part 1. Numerical verification of the proposals is presented in Part 4.

## Conclusions

1. Experimental evidence shows that superplasticity in Zn-22 wt-%Al alloy is the result of interconnected cooperative sliding along localised shear surfaces situated throughout the volume of the specimen. This shear is accomplished by cooperative grain/interphase boundary sliding involving a number of boundaries that slide together as an entity. On the specimen surface cooperative boundary sliding manifests itself as wavy, localised shear bands.

2. Along boundaries that give rise to cooperative grain/interphase boundary sliding, triple junctions move to nearly 180° to orient themselves favourably to the shear stresses that result from the external load. Boundaries that are conducive to easy sliding, e.g. high angle boundaries, boundaries of large misfit ('more open boundaries'), and interphase interfaces in the Zn-22 wt-%Al alloy, participate preferentially in the cooperative grain/interphase boundary sliding process.

3. When plane interface formation involves an interphase interface, cooperative sliding is accommodated by dislocation motion through the lattice and this provides direct experimental support for the model proposed in Part 1.<sup>12</sup>

4. Cooperative grain/interphase boundary sliding results in an increased number of non-equilibrium triple junctions that meet at angles significantly different from 120°. This observation is indicative of local boundary migration (caused by diffusion at intercrystalline boundaries and lattice dislocation motion at interphase interfaces) being present as a process that accommodates boundary sliding.

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